

| Question |     |      |  | Marking details                                                                                                                                                                        | Marks available |          |          |          |          |          |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                        | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 3        | (a) | (i)  |  | Increase (1)<br>Momentum (1)<br>Decreases (1)                                                                                                                                          | 3               |          |          | 3        |          |          |
|          |     | (ii) |  | Crumple zone / seat belt / roll cage / side impact bars /<br>child safety seat / head rests<br>Accept child locks<br>Don't accept any reference to brakes / tyres                      | 1               |          |          | 1        |          |          |
|          | (b) |      |  | This reduces {air resistance / drag} (1) Don't accept wind resistance<br>Making the car [more energy] efficient <b>or</b> to reduce fuel costs (1)<br>Accept allows the car to go fast | 2               |          |          | 2        |          |          |
|          |     |      |  | <b>Question 3 total</b>                                                                                                                                                                | <b>6</b>        | <b>0</b> | <b>0</b> | <b>6</b> | <b>0</b> | <b>0</b> |